

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. Local artificial intelligence (AI) can process traffic information closer to the road and support faster decisions. This paper presents a conceptual smart traffic-light architecture using local AI, including street cameras, smart pole computers, a city cloud server, and traffic lights. The proposed system focuses on processing traffic information locally while forwarding selected information to the cloud for storage and analysis.

Index Terms—Smart traffic lights, Local AI, Artificial Intelligence, Traffic Signal Control, Machine Vision, Edge Computing

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. Instead of changing lights only according to a clock, the system uses information collected from the road.

Research reviews show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. CLOUD AI AND LOCAL AI

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions.

A small local computer can examine camera input and send a short instruction to the traffic-light controller.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

III. PROPOSED SYSTEM

The proposed system uses local AI to process traffic information near the traffic signal. The major components of the system are:

- 1) Street Cameras
- 2) Smart Pole Computers

3) City Cloud Server

4) Traffic Lights

The cameras capture traffic conditions. The smart pole computer processes the camera information locally and makes a quick traffic decision. Selected information can then be forwarded to the city cloud server for storage and wider analysis.

Finally, the decision is sent to the traffic lights, which change their signals according to the traffic condition.

IV. SYSTEM ARCHITECTURE

The proposed architecture follows four major steps:

- 1) Street Cameras capture traffic conditions.
- 2) Smart Pole Computers perform local processing and make a quick traffic decision.
- 3) Selected information can be forwarded to the City Cloud Server for storage and wider analysis.
- 4) The decision is sent to the Traffic Lights, which change their signals.

The overall flow is:

A. System Flowchart

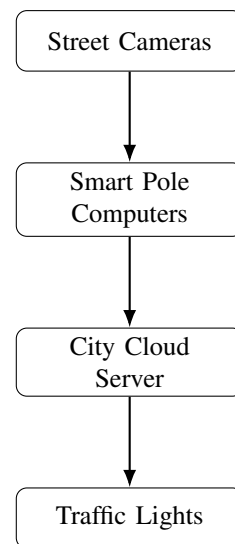


Fig. 1. Proposed smart traffic-light system architecture.

V. SYSTEM COMPARISON

The systems can be compared using three simple measures: decision-making time, internet data usage, and setup cost.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30 seconds	Zero	\$200
Cloud AI Systems	2 seconds	High	\$2,500
Local Smart Cameras	0.5 seconds	Low	\$1,200

Table I provides a simple conceptual comparison of the three systems.

VI. PERFORMANCE MEASURES

Three basic measures can be used to compare traffic-light systems.

A. Decision Time

Decision time represents how quickly the system can respond to changing traffic conditions.

$$T_d = T_p + T_c \quad (1)$$

where

- T_d is the total decision time,
- T_p is the processing time,
- T_c is the communication time.

For a local AI system, communication requirements can be reduced because much of the processing takes place near the traffic signal.

B. Internet Data Usage

Local processing can reduce the amount of camera information that needs to be transmitted to a remote server.

The approximate transmitted data can be represented as:

$$D_t = R_v \times T \quad (2)$$

where

- D_t is transmitted data,
- R_v is the video data rate,
- T is the transmission time.

Processing information locally can reduce the amount of raw video data transmitted over the network.

VII. ADVANTAGES OF LOCAL AI

The proposed local AI approach has several potential advantages:

- Faster local decision making.
- Reduced dependence on continuous internet connectivity.
- Lower amount of raw camera data transmitted to the cloud.
- Ability to process traffic information close to the road.
- Support for adaptive traffic signal control.

VIII. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable.

However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

IX. FUTURE WORK

Future development of the proposed system can focus on emergency vehicle detection and signal coordination.

An AI-based system could detect an approaching emergency vehicle and communicate with nearby traffic signals. The signals could then be coordinated to provide a clear route.

Future research can also investigate:

- Emergency vehicle detection using computer vision.
- Real-time traffic-density estimation.
- Adaptive signal timing.
- Edge AI hardware for traffic applications.
- Communication between nearby traffic signals.
- Evaluation using real-world traffic datasets.

X. CONCLUSION

Smart traffic lights can use camera information and artificial intelligence to respond to changing traffic conditions.

Traditional timed systems are simple, but they have limited ability to react to sudden traffic changes. Cloud-based AI can provide centralized processing but may depend on internet connectivity and can introduce communication delay.

Local AI provides an approach in which traffic information can be processed closer to the road. The proposed architecture combines street cameras, smart pole computers, a city cloud server, and traffic lights.

A further research opportunity is emergency vehicle detection and automatic coordination of traffic signals to create a clear route for ambulances.

ACKNOWLEDGMENT

The authors would like to thank the faculty and department for providing guidance and support for this research work.

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